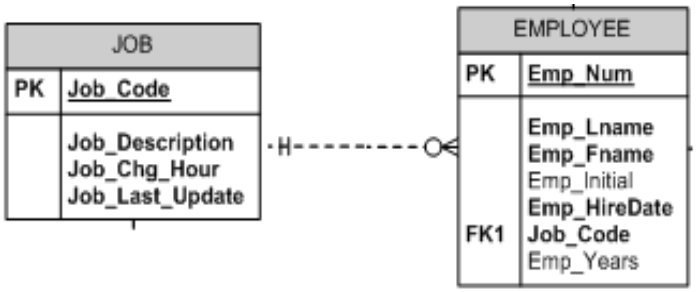
Tutorial 9 - Guide
Language Integrated Query (LINQ)

Tutorial Objectives:

- **Learn LINQ:**
  - Write C# statements using LINQ syntax

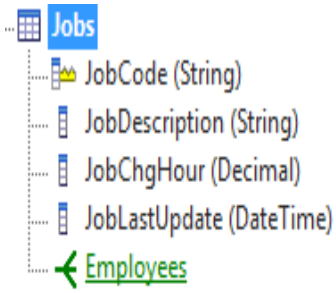
- **Tutorial Tasks:**

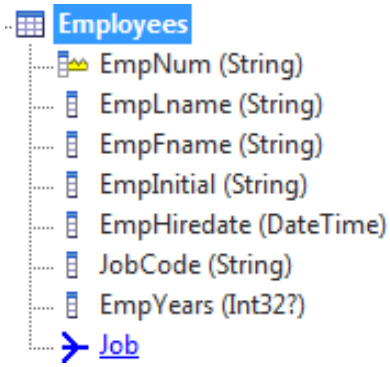
  Consider the following two tables from the database used in Tutorial 5.



a. If you have dropped or altered the above tables, run the script Tutorial9DBScript.txt provided on Stream to create and populate the tables again using SQL Developer.

b. Download LINQPad5 from the https://www.linqpad.net/ and follow the instructions from Week9 lecture slides to run the application and make a connection to the Oracle database. Both these tables should be available to you as shown:





c. Write the following 10 statements in C# using LINQ syntax:

Packed with many code examples, week 9 lecture materials provide most of the guidance needed to get started and finish this tutorial. The materials guide the students with no/minimal C# experience. Will be good to first write and run the code from the lecture slides' examples. The learning curve for some may be bit steep but is all doable. The learning will help both the tutorial work and assignment LINQ component (11-12% of the Part B assignment).

When you write LINQ code unlike SQL, names are case sensitive here (e.g., attribute JobCode will work but jobcode won't work; Table name Jobs will work, and jobs won't; constants declared, etc). You would have already noticed the change in the table names (which are plural – e.g., Jobs – because of the option we chose while doing connection configuration).

When you run code make a note of the following:

- you are connected to the Oracle server (assuming you have added the connection already) when you run your code. Under LINQ, Connection dropdown has <None> as the default connection option and you need to select your connection.
- you have selected the right option under Language dropdown (e.g., C# Statement(s)). The default is C# Expression. Your lecture examples have demonstrated options such as C# Program and C# Statements. For tutorial work most of you will write statements (or a program).
- your query results require some method to display [e.g., myQuery.Dump();]
- end of statement character semi-colon [;] after each statement.

Use the given query hints, only when stuck. There are multiple ways to write the code. Once you write a couple of queries your code will start to flow 😊

1. Display all jobs having jobChgHour greater than 25. The list should be in descending order for job charge hour rate.

   Could declare job rate value (25) as a decimal value (see slide 31 for different options for numeric values).

2. Display all jobs which were updated before 20/12/2018 date.

   Use C# syntax - DateTime DT = Convert.ToDateTime("20/12/2018");

   Basically, need to convert string to date type for comparison to be valid (as attribute JobLastUpdate is of the type DateTime).

3. Display all employees who were hired after 20/12/1999. This list should be ordered in ascending order for the hire date.

4. Display all employees who work as Systems Analyst.

   Comparison operator is == (e.g., course == "Databases")

5. Display all employees who have the word “Engineer” in their job descriptions.

   You will need to join two tables (slide 40).

   A method called Contains may be used along with JobDescription(“something”).

6. Display the name of the employee who was hired first.

   Slide 33 talks about aggregates. So, need to get the hire date for the employee who has been working the longest (was hired first) i.e., Min() value of the employee hire date.

7. Display all employees who have a value for EmpInitial (i.e., is not null)

   ! can be used for negation here.

8. Display all employees for whom first name starts with D.

   Your lecture slide has a similar example (e.g., names with A)

9. Display the total number of employees in each job code.

   This one and the 10th query require group operator (slides 43-46).

10. Display all employees who have been in a job position for less than 22 years.

    This query can be written using employee years (EmpYears) attribute. Not required to calculate using the current date.